



National youth Services Council
Information Security and Ethical Hacking
2026 January – June
Final Examination
Practical Exam



Answer all the questions. (100 Marks)

Time: - 3 hours

Brindarri Company is a reputable marketing company in Sri Lanka. Over the past few years, the company has developed its business using a workgroup-based network infrastructure. With the rapid growth experienced in recent years, the company has now planned to implement proper security policies and technologies and to migrate from the existing workgroup network infrastructure to a domain-based client network infrastructure. Currently, the company has more than **350 staff members across 10 departments**.

Network Infrastructure Details:

- Windows Server 2022 **configured as an** Active Directory Domain Controller (AD DC) **with** DHCP **and** DNS services enabled.
- 350 client computers (**for practical implementation, you can create and test with one** client computer).
- Kali Linux Server **and** Ubuntu Server.
- IP Segment: **192.168.100.0/24**

1. Draw a proper network diagram for the above infrastructure. Then **list the steps** to create this lab in a VMware environment (e.g., create VMs, assign IP addresses, configure network adapters).

(10 marks)

2. List and explain **three main categories of cybersecurity controls** (e.g., technical, administrative, physical).

(10 marks)

3. Cloud providers use data centers to offer data storage solutions to their customers. Cloud storage allows data to be accessed by any authorized user with internet access. Describe **three measures** you could take to secure data when using cloud storage.

(10 marks)

4. Using **Event Viewer** or **PowerShell (Get-WinEvent)** on Windows Server, filter and display the following system event IDs: **41, 1100, 1102, 6008**. Provide the exact filter method (GUI steps or PowerShell command) and a screenshot for each event.

(15 marks)

Event ID	Description
41	The system rebooted without a clean shutdown (Kernel Power event, often hardware related).
1100	The event logging service has shut down.
1102	Audit log was cleared (potentially suspicious activity).
6008	The system shut down unexpectedly.

5. Install Ubuntu Server (provided as a VM). Using the terminal, perform the following tasks:
(10 marks)

- Reset the root password using recovery mode.
- Create a directory named `security_lab` in your home folder and check its disk usage (e.g., `du -sh`). (
- List all files and directories (including hidden) inside `/var/log`.

6. How do you update the Kali Linux OS and check for available updates? Provide the exact commands.
(5 marks)

7. Find the DNS record information (A, MX, NS records) for the following websites:
`www.google.com` and `www.itlanka.edu.lk`
Provide the command used (e.g., `dig` or `nslookup`) and the output for each domain.
(10 marks)

8. Using **Wireshark** or **tcpdump** on Kali Linux, monitor data packets passing through the network adapter and filter **HTTPS-related** packet information (TCP port 443). Provide the filter expression and a screenshot of captured packets.
(10 marks)

9. Name **two free website vulnerability scanning tools** (e.g., OWASP ZAP, Nikto). Choose **one** tool and explain how to use it to scan **your own lab web server** (simulated – do not scan external sites without permission). Describe the steps and what information you would look for to identify vulnerabilities.
(10 marks)

10. Using Kali Linux, perform an **authorised password attack** against the Ubuntu server's SSH login. Name the tool (e.g., Hydra) and provide the exact command.

Disclaimer: Only attack lab machines that you own or have explicit permission to test. Unauthorised attacks are illegal.
(10 marks)